Yong Gao — Curriculum Vitae

Contact Information

0.27, Albert Einstein Institute
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Education

Postdoctoral researcher, Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam, Germany **September 2023–Present**
Computational Relativistic Astrophysics division, led by Prof. Masaru Shibata

Ph.D., Physics, Peking University, Beijing, China **August 2018–August 2023**

Thesis Advisor: Prof. Lijing Shao

Thesis Title: *Study of neutron-star structures and spin precession in the era of multimessenger astrophysics*

B.S., Physics, Dalian University of Technology, Dalian, Liaoning Province, China **July 2018**

Degree conferred with honor

Senior Dissertation Advisors: Prof. Renxin Xu and Prof. Chong Li

Dissertation Title: *The electron distributions of strangelets in the Thomas-Fermi model*

Research Interests

Dense matter and relativistic hydrodynamics of neutron stars. Structure and universal relations; oscillations and instabilities of rotating neutron stars; dynamical tides; numerical simulations of binary neutron-star mergers and waveform modelling.

Testing strong-field gravity. Neutron-star structure in alternative theories of gravity; numerical simulations of binary mergers in alternative theories of gravity; waveform modelling in scalar-tensor gravity.

Honors and Awards

Principal Scholarship, Peking University **2022-2023**

Tung Scholarship, Peking University **2021–2022**

Merit Student, Peking University **2021–2022**

The Second Prize for Oral Presentation, Physics Five Universities **April 2021**

Vela Prize for Oral Presentation, FAST/Future Pulsar Symposium 9 **August 2020**

National Scholarship, Peking University **2019–2020**

Excellent Teaching Assistant Award, Peking University **2019–2020**

Principal Scholarship, Peking University **2018–2019**

Learning Excellence Award (First Prize), Dalian University of Technology **2015–2016**

Teaching Experience

Lecturer, Max Planck Institute for Gravitational Physics

Jürgen Ehlers Spring School: Neutron stars and gravitational waves **Spring 2026**

Jürgen Ehlers Spring School: Introduction to neutron-star physics **Spring 2025**

Teaching Assistant, Peking University

Electrodynamics (B), **Excellent Teaching Assistant Award** **Fall 2022**

General Physics I, Mechanics & Electromagnetism **Fall 2021**

Theoretical Mechanics (A), **Excellent Teaching Assistant Award** **Fall 2019**

Co-advised Students

Undergraduate Students, Peking University

Haoyang Qi, co-advised with Prof. Lijing Shao **2021-2022**
Constraints on ultralight dark matter with pulsar timing

Huimei Wang, co-advised with Prof. Lijing Shao **2020-2021**
Undergraduate Dissertation: The structure of neutron stars with anisotropic pressure

Jingyuan Deng, co-advised with Prof. Lijing Shao **2020-2021**
Undergraduate Dissertation: Forced precession of neutron stars

Zexin Hu, co-advised with Prof. Lijing Shao **2020-2021**
Scalarized neutron stars in massive scalar-tensor gravity

Computer Skills

Proficient in Mathematica, Julia and Python. Experience in C++, Fortran, Bash and HPC.
Numerical methods: relativistic stellar perturbations, spectral and eigenvalue calculations, binary initial data with FUKA, and numerical-relativity simulations with SACRA-K.
Markup languages: LaTeX, Markdown.

Code development— https://github.com/GravYong.

Professional Activities, Outreach, and Service

KAGRA Collaboration

Member of KAGRA Future Strategy Committee (FSC) **2021–2023**

Chair of conference session/group meeting

KAGRA Future Working Group 1st Open Meeting (*online*) **November 2021**

Chair of the Kiaagravity group meeting **2020-2021**

Journal referee

Physical Review D (PRD) **2024–Present**

Classical and Quantum Gravity (CQG) **2021-Present**

Research in Astronomy and Astrophysics (RAA) **2021-Present**

Science China Physics, Mechanics & Astronomy (SCPMA) **2021-Present**

Submitted Papers and Preprints

3. **Y. Gao** and J. Steinhoff. *Relativistic tidal response and mode excitation in neutron-star binaries.* (2026). [arXiv:2611.14243].
2. **Y. Gao** and M. Shibata. *Quasi-normal Modes and Instabilities of Rotating Relativistic Stars.* (2026). [arXiv:2611.03986].
1. **Y. Gao**, M.-Z. Han, K. Kiuchi, M. Shibata, E. Zhou, and K. Hotokezaka. *Subsolar-mass binary mergers of strange stars and neutron stars: gravitational waves and ejecta.* (2026). [arXiv:2607.07668].

Refereed and Accepted Journal Articles

35. H. Shen, **Y. Gao**, H.-B. Li, and R.-X. Xu. *Tidal Deformation and Strain Accumulation of Solid Compact Stars*. Mon. Not. R. Astron. Soc., stag2059 (2026; accepted manuscript published online). [arXiv:2607.11780].
34. H. Sun, J.-X. Niu, **Y. Gao**, H.-B. Li, and C.-J. Xia. *Asteroseismology of neutron stars with both hyperons and Δ resonances*. Phys. Rev. D (accepted, 2026). [arXiv:2606.24343].
33. D. Gao, H.-J. Kuan, Y. Zhou, Z. Miao, M.-Z. Han, **Y. Gao**, C. Zhang, and E. Zhou. *Difference between quark stars and neutron stars in universal relations and their effect on gravitational waves*. Phys. Rev. D **114**, 043058 (2026). [arXiv:2512.24703].
32. S. Li, H. Lü, **Y. Gao**, R. Xu, L. Shao, and H. Yu. *Neutron stars more compact than black holes as a probe of strong-field gravity*. Phys. Rev. D **114**, L021504 (2026). [arXiv:2312.01406].
31. Z. Hu, **Y. Gao**, and L. Shao. *Linear analysis of I-C-Love universal relations for neutron stars*. Phys. Rev. D **113**, 044056 (2026). [arXiv:2505.13110].
30. **Y. Gao**, K. Hayashi, K. Kiuchi, A. T.-L. Lam, H.-J. Kuan, and M. Shibata. *Convective stability analysis of massive neutron stars formed in binary mergers*. Phys. Rev. D **113**, 023011 (2026). [arXiv:2501.19053].
29. **Y. Gao**, H.-J. Kuan, C.-J. Xia, H. O. Silva, and M. Shibata. *Nonradial oscillations of stratified neutron stars with solid crusts: Mode characterization and tidal resonances in coalescing binaries*. Phys. Rev. D **112**, 123006 (2025). [arXiv:2509.00257].
28. S.-X. Yi, W. Zhao, R.-X. Xu, X.-F. Wu, G. Stratta, S. Dall'Osso, Y.-J. Xu, A. Santangelo, S. Zane, S.-N. Zhang, et al. (including **Y. Gao**). *Prospects for time-domain and multi-messenger science with eXTP*. Sci. China Phys. Mech. Astron. **68**, 119506 (2025). [arXiv:2506.08368].
27. M.-Z. Han, **Y. Gao**, K. Kiuchi, and M. Shibata. *Dependence of postmerger properties on the thermal heating efficiency in neutron star mergers*. Phys. Rev. D **112**, 023005 (2025). [arXiv:2504.08514].
26. A. T.-L. Lam, **Y. Gao**, H.-J. Kuan, M. Shibata, K. Van Aelst, and K. Kiuchi. *Accessing Universal Relations of Binary Neutron Star Waveforms in Massive Scalar-Tensor Theory*. Phys. Rev. Lett. **134**, 151402 (2025). [arXiv:2410.00137].
25. Z. Wang, **Y. Gao**, D. Liang, J. Zhao, and L. Shao. *Vetting quark-star models with gravitational waves in the hierarchical Bayesian framework*. JCAP **11**, 038 (2024). [arXiv:2409.11103].
24. Y. Liu, H.-B. Li, **Y. Gao**, L. Shao, and Z. Hu. *Effects from dark matter halos on x-ray pulsar pulse profiles*. Phys. Rev. D **110**, 083018 (2024). [arXiv:2408.04425].
23. Y. Kang, C. Liu, J.-P. Zhu, **Y. Gao**, L. Shao, B. Zhang, H. Sun, Y.-H. I. Yin, and B.-B. Zhang. *Prospects for detecting neutron star–white dwarf mergers with decihertz gravitational-wave observatories*. Mon. Not. R. Astron. Soc. **528**, 5309 (2024). [arXiv:2309.16991].
22. G. Yim, **Y. Gao**, Y. Kang, L. Shao, and R. Xu. *Continuous gravitational waves from trapped magnetar ejecta and the connection to glitches and antiglitches*. Mon. Not. R. Astron. Soc. **527**, 2379–2392 (2024). [arXiv:2308.01588].
21. S. Chen, **Y. Gao**, E. Zhou, and R. Xu. *Free Energy of Anisotropic Strangeon Stars*. Res. Astron. Astrophys. **24**, 025005 (2024). [arXiv:2305.19687].
20. E. Zhou, **Y. Gao**, Y. Zhou, X. Lai, L. Shao, W. Wang, S. Xiong, R. Xu, S. Yi, G. Yim, et al. *The Precursor of GRB211211A: A Tide-induced Giant Quake?*. Res. Astron. Astrophys. **24**, 025019 (2024). [arXiv:2305.10682].
19. G. Desvignes, P. Weltevrede, **Y. Gao**, D. I. Jones, M. Kramer, M. Caleb, R. Karuppusamy, L. Levin, K. Liu, A. G. Lyne, L. Shao, and B. Stappers. *A freely precessing magnetar following an X-ray outburst*. Nature Astronomy **8**, 617–627 (2024).
18. H.-B. Li, **Y. Gao**, L. Shao, and R.-X. Xu. *Asteroseismology of Compact Stars*. Universe **10**, 157 (2024).

17. C. Zhang, **Y. Gao**, C.-J. Xia, and R. Xu. *Hybrid strangeon stars*. Phys. Rev. D **108**, 123031 (2023). [arXiv:2309.14114].
16. H. Liu, **Y. Gao**, Z. Li, A. Dohi, W. Wang, G. Lv, and R. Xu. *EOS-dependent millihertz quasi-periodic oscillation in low-mass X-ray binary*. Mon. Not. R. Astron. Soc. **525**, 2054 (2023). [arXiv:2308.05288].
15. C. Zhang, **Y. Gao**, C.-J. Xia, and R. Xu. *Rescaling strange-cluster stars and its implications on gravitational-wave echoes*. Phys. Rev. D **108**, 063002 (2023). [arXiv:2305.13323].
14. **Y. Gao**, L. Shao, and J. Steinhoff. *A Tight Universal Relation between the Shape Eccentricity and the Moment of Inertia for Rotating Neutron Stars*. Astrophys. J. **954**, 16 (2023). [arXiv:2303.14130].
13. H.-B. Li, **Y. Gao**, L. Shao, R. Xu, and R.-X. Xu. *g-mode of neutron stars in pseudo-Newtonian gravity*. Phys. Rev. D **108**, 064005 (2023). [arXiv:2302.03856].
12. **Y. Gao**, L. Shao, G. Desvignes, D. I. Jones, M. Kramer, and G. Yim. *Precession of magnetars: dynamical evolutions and modulations on polarized electromagnetic waves*. Mon. Not. R. Astron. Soc. **519**, 1080 (2023). [arXiv:2211.17087].
11. H.-B. Li, **Y. Gao**, L. Shao, R.-X. Xu, and R. Xu. *Oscillation modes and gravitational waves from strangeon stars*. Mon. Not. R. Astron. Soc. **516**, 6172 (2022). [arXiv:2206.09407].
10. R. Xu, **Y. Gao**, and L. Shao. *Neutron stars in massive scalar-Gauss-Bonnet gravity: Spherical structure and time-independent perturbations*. Phys. Rev. D **105**, 024003 (2022). [arXiv:2111.06561].
9. **Y. Gao**, X.-Y. Lai, L. Shao, and R.-X. Xu. *Rotation and deformation of strangeon stars in the Lennard-Jones model*. Mon. Not. R. Astron. Soc. **509**, 2758 (2022). [arXiv:2109.13234].
8. Z. Hu, **Y. Gao**, R. Xu, and L. Shao. *Scalarized neutron stars in massive scalar-tensor gravity: X-ray pulsars and tidal deformability*. Phys. Rev. D **104**, 104014 (2021). [arXiv:2109.13453].
7. J. Zhao, L. Shao, **Y. Gao**, C. Liu, Z. Cao, and B.-Q. Ma. *Probing dipole radiation from binary neutron stars with ground-based laser-interferometer and atom-interferometer gravitational-wave observatories*. Phys. Rev. D **104**, 084008 (2021). [arXiv:2106.04883].
6. R. Xu, **Y. Gao**, and L. Shao. *Signatures of Lorentz Violation in Continuous Gravitational-Wave Spectra of Ellipsoidal Neutron Stars*. Galaxies **9**, 12 (2021). [arXiv:2101.09431].
5. R. Xu, **Y. Gao**, and L. Shao. *Precession of spheroids under Lorentz violation and observational consequences for neutron stars*. Phys. Rev. D **103**, 084028 (2021). [arXiv:2012.01320].
4. **Y. Gao** and L. Shao. *Precession of triaxially deformed neutron stars*. Astron. Nachr. **342**, 364 (2021). [arXiv:2011.04472].
3. R. Xu, **Y. Gao**, and L. Shao. *Strong-field effects in massive scalar-tensor gravity for slowly spinning neutron stars and application to X-ray pulsar pulse profiles*. Phys. Rev. D **102**, 064057 (2020). [arXiv:2007.10080].
2. **Y. Gao**, L. Shao, R. Xu, L. Sun, C. Liu, and R.-X. Xu. *Triaxially-deformed Freely-precessing Neutron Stars: Continuous electromagnetic and gravitational radiation*. Mon. Not. R. Astron. Soc. **498**, 1826 (2020). [arXiv:2007.02528].
1. C. Liu, L. Shao, J. Zhao, and **Y. Gao**. *Multiband Observation of LIGO/Virgo Binary Black Hole Mergers in the Gravitational-wave Transient Catalog GWTC-1*. Mon. Not. R. Astron. Soc. **496**, 182 (2020). [arXiv:2004.12096].

Conference Proceedings

1. **Y. Gao**, R. Xu, and L. Shao. *Precession of Neutron Stars Under Lorentz Violation and Observational Consequences*. Proceedings of the Ninth Meeting on CPT and Lorentz Symmetry (2023).

Popular Science Articles

4. **Y. Gao**, G. Desvignes, L. Shao, (2024) A freely precessing magnetar (*in Chinese*).
3. **Y. Gao**, L. Shao, R.-X. Xu, (2019) The waltz of a binary neutron star system (an article about GW170817, *in Chinese*).
2. **Y. Gao**, (2022) The structures of neutron stars (an article about dense matter in neutron stars, *in Chinese*).
1. **Y. Gao**, L. Shao, (2022) Does Einstein's theory of gravity hold up to the latest LIGO/VIRGO/KAGRA observations? (**translated** from the English version).

Invited Talks

7. University of Tübingen, TAT Einstein Seminar — October 2026
6. Yangzhou University, School of Physics Science and Technology, Seminar — December 2024
5. Dalian University of Technology, Seminar — July 2023
4. Yangzhou University, School of Physics Science and Technology, Seminar — September 2022
3. Peking University, School of Physics, CuiYing Graduate Student Salon — February 2021
2. Max Planck Institut für Gravitationsphysik Colloquium (*online*) — September 2020
1. University of Tartu, Theoretical Physics Laboratory Colloquium (*online*) — October 2020

Contributed Talks

12. YITP workshop: Multi-Messenger Astrophysics in the Dynamic Universe
Kyoto, Japan — February 2026
11. Workshop on Gravitational Waves and Compact Stars
Purple Mountain Observatory — January 2025
10. Brainstorming workshop: Deciphering the equation of state using gravitational waves from astrophysical sources, University of Warsaw, Poland — August 2024
9. SKA Pulsar Science Symposium 2022 — August 2022
8. FAST/Future Pulsar Symposium 11 — August 2022
7. Summer Science Day, KIAA, Peking University — July 2022
6. The 60th Anniversary of X-Ray Astronomy (*online*) — June 2022
5. Ninth Meeting on CPT and Lorentz Symmetry (*online*) — May 2022
4. FAST/Future Pulsar Symposium 10 — July 2021
3. Gravitation and Relativistic Astrophysics, Chinese Physical Society — April 2021
2. Gravitation and Cosmology Symposium — December 2020
1. FAST/Future Pulsar Symposium 9 — August 2020

Posters

2. General-relativistic simulations of binary strange star mergers
CSQCD2026, Barcelona, Spain — May 2026
1. Convective stability analysis of massive neutron stars formed in binary mergers
New Frontiers in NR 2025, Palma de Mallorca, Spain — July 2025